D Simran Patra

Phone: +91 8144904936 | Email: simranpatra754@gmail.com | Location: Berhampur, Ganjam, Odisha
LinkedIn: linkedin.com/in/d-simran-patra-b53684340/ | GitHub: github.com/simran21122

PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in frontend and backend web development, including building RESTful APIs during a software development internship. Worked on projects including an LLM-powered assistant, a deployed content-based recommendation system and a vector database with a RAG pipeline. Strong problem-solver and quick learner, eager to contribute to a collaborative engineering team.

EDUCATION

NIST University CGPA: 8.9
2023 – 2027 *B.Tech. in Computer Science and Engineering*

Sashi Bhusan Rath Government Women's H.S. 77%
2021 – 2023 *Intermediate (12th)*

Sree Girls High School, Berhampur 84.66%
Matriculation (10th)

INTERNSHIP EXPERIENCE

Risoapp India Pvt Ltd Berhampur
May 2026 – Aug 2026 *Software Developer Intern*

- Contributed to a production-level application by building and maintaining RESTful backend APIs and database operations using Node.js and Express.js
- Collaborated with the engineering team to integrate React.js frontend components with backend services, gaining practical experience with Git-based version control workflows

Central Tool Room & Training Centre (MSME), Govt. of India Bhubaneswar
Jun 2025 – Jul 2025 *AI-ML Intern*

- Built a real-time Face Recognition System using OpenCV, CNN, and Keras, applying deep learning, NLP, and computer vision concepts learned during structured AI/ML training

SKILLS

Programming Languages: Java, Python, JavaScript
Web Technologies: React.js, Node.js, Express.js, HTML5, CSS3, Tailwind CSS
AI / ML: Machine Learning, NLP, TF-IDF, scikit-learn, RAG, Vector Databases, LLM Integration
Databases & Backend: SQL, MongoDB, REST API Design
Core CS Concepts: Data Structures, Algorithms, Object-Oriented Programming (OOP), Database Management (DBMS)
Tools & Platforms: GitHub, Git, Postman, VS Code, Figma, Canva

PROJECTS

CREDEX.IO – Measure What You Can Build (Team Project | Frontend Developer)
GitHub: github.com/simran21122/CREDEX | Live Demo: credex-io.vercel.app

- Developed the React.js frontend for a developer credibility scoring platform, building responsive score card, ATS checker and roadmap builder pages with Tailwind CSS

SimranOS: AI-Powered Portfolio (Solo Project)
GitHub: github.com/simran21122/AI_Portfolio | Live Demo: simran-ai-portfolio.onrender.com

- Engineered an LLM-powered portfolio assistant with Python, FastAPI and the Groq API and deployed it on Render; used prompt engineering to limit hallucination and prompt injection, with real-time streaming over Server-Sent Events and multi-turn memory
- Secured the app with Pydantic request validation, sanitized output rendering and environment-based API key management, and delivered a responsive JavaScript interface

VectorDB Studio – Vector Database and RAG Pipeline (Solo Project)
GitHub: github.com/simran21122/AI_Model

- Implemented a vector database from scratch in Python with HNSW, KD-Tree and Brute Force search, three distance metrics and a latency benchmark
- Created a RAG pipeline that chunks documents, generates 768-D embeddings and answers questions using Gemini API or Ollama

CineMatch – Movie Recommendation System (Solo Project)
GitHub: github.com/simran21122/CINEMATCH | Live Demo: cinematch-f9e1.onrender.com

- Deployed a movie recommendation web application on Render with an HTML, CSS and JavaScript frontend and a Python REST API backend
- Applied a TF-IDF similarity model for content-based recommendations and the TMDB API for genre-based recommendations

KEY ACHIEVEMENTS & CERTIFICATIONS

Runner-up – Hack Arena 2.0 National-Level Hackathon *IEEE Student Branch, Guru Nanak Institutions, Hyderabad*

- Placed second among 40+ participating teams

3rd Position – Code Kriti 4.0 Innovation Challenge (Mar 2026) *Parala Maharaja Engineering College*

- Analyzed a failed startup case study and presented improvement strategies in the qualifier round, then pitched an original startup idea with Q&A in the final round
- **Artificial Intelligence** – IBM (Feb 2026)
- **Digital Skills: User Experience (92%)** – Accenture x FutureLearn (Oct 2024)
- **AWS Cloud with AI** – LinuxWorld National-Level (Jan 2024)